

Drafting customs declarations 4x faster, with a human always in the loop

Hekkers Customs Brokerage, Rotterdam · Engagement Q2 2025 to ongoing

Industry	Customs brokerage, Port of Rotterdam, mid-size (28 staff)
Service line	Agentic workflow — HS-code suggestion + duty draft
Engagement length	10-week build, 11 months in production
Investment	€14,200 build + €1,100/month managed
Scope	Inbound declaration request → draft HS code + duty calc → broker review

The Challenge

Hekkers handles roughly 380 customs declarations per week across import and export desks. The single most expert-intensive task is HS-code classification: deciding which 8- or 10-digit Combined Nomenclature code applies to a given shipment, based on commercial invoices and product descriptions. Senior brokers spent 12 to 18 minutes per declaration on this step alone, and the resulting bottleneck meant junior staff were underutilised — they could draft the simpler declarations but not the ambiguous ones.

Misclassification has real consequences: an underpaid duty can trigger a customs audit; an overpaid duty is money the importer loses. The operations director, Joris Hekkers, was clear with us from the start: this engagement would not pass internal review if the agent classified declarations without senior-broker approval. The role of the agent was to do the preparation work, never to commit a code.

Our Approach

We built a three-step workflow. Step one extracts structured shipment data from incoming commercial invoices and packing lists, including PDFs and forwarded carrier emails. Step two queries a vector index of Hekkers' own historical declarations (roughly 38,000 past classifications, anonymised and embedded) to surface the five most-similar past shipments and the codes they were classified under. Step three drafts a recommended HS code with confidence score, alternative codes the broker should consider, and a duty calculation against the current EU TARIC rates.

The output is presented in a custom interface that sits beside the broker's primary system (Descartes). Brokers see the draft, the supporting historical examples, the confidence score, and any flags — for example, “commodity description mentions lithium-ion, dangerous-goods classification may also apply.” The broker accepts, edits, or overrides. Every override is logged and feeds back into the next nightly index refresh, so the system learns from senior-broker corrections.

We deliberately did not connect the agent to Descartes for write operations. The final declaration is keyed in by a human, every time. Hekkers' compliance team requested this constraint in week two, and we have kept it ever since — even though it would be technically straightforward to remove.

Results

4.1x	87%	0	+€340k
faster median time-per-declaration	first-pass broker acceptance rate	compliance incidents in 11 months	annualised broker capacity recovered

The 87% first-pass acceptance figure is the metric Hekkers tracks weekly. It started at 71% in the first month and improved as the feedback loop from broker corrections fed back into the historical index. The remaining 13% break down roughly evenly between minor edits (typically duty rate updates that lagged the TARIC refresh) and full overrides (genuinely ambiguous shipments where the senior broker disagreed with the surfaced precedent).

In the client's words

“The thing that surprised me was how quickly the junior brokers became more confident. They could see, on every declaration, what a senior would probably have done and why. Six months in, our junior team is independently handling declaration types that used to require senior review. The agent was supposed to save senior time, and it did — but the bigger win was the training effect.”

Joris Hekkers, Operations Director — Hekkers Customs Brokerage

Lessons Learned

- Agentic workflows in regulated domains must be drafting tools, not autonomous actors. The compliance team should explicitly approve the human-in-the-loop boundary before the build starts.
- Historical-data retrieval beats general LLM reasoning for domain-specific classification. We did not need a fine-tuned model; we needed Hekkers' own past work in a vector index.
- Override logging is the underrated feature. The system improves weekly because senior corrections feed back into retrieval — without that loop, accuracy would have plateaued at month two.
- The training-effect benefit was unplanned and ended up larger than the time-savings benefit. We now ask about junior-staff development in discovery for similar engagements.